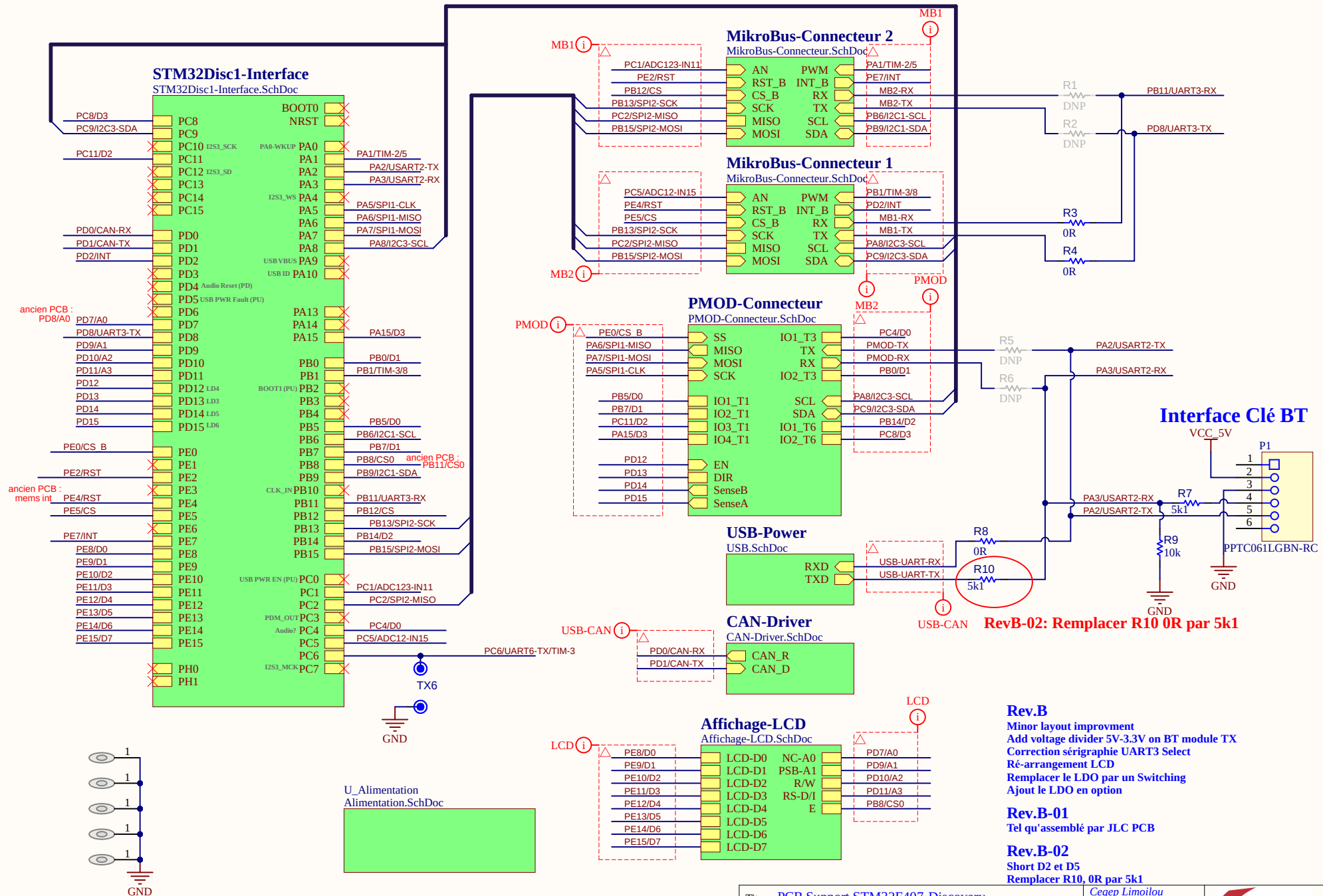


PCB Support STM32F407-Discovery



Rev.B

- Minor layout improvement
- Add voltage divider 5V-3.3V on BT module TX
- Correction sérigraphie UART3 Select
- Ré-arrangement LCD
- Remplacer le LDO par un Switching
- Ajout le LDO en option

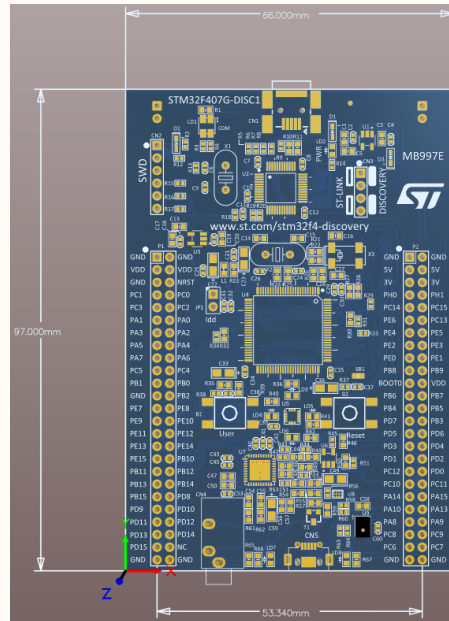
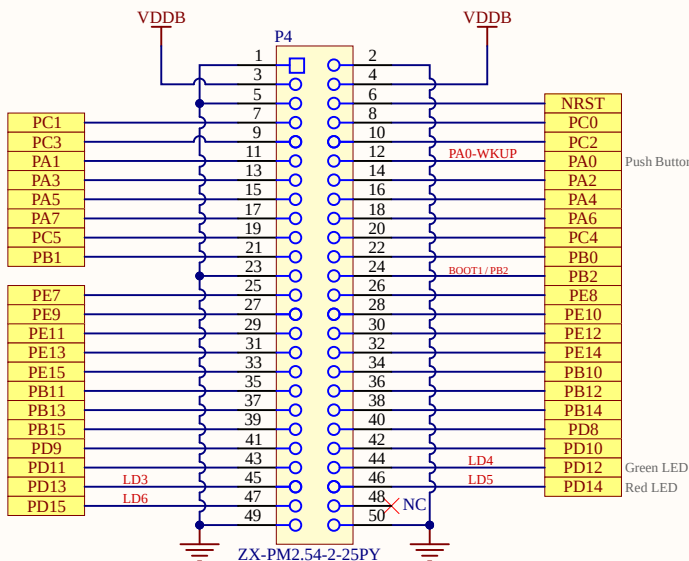
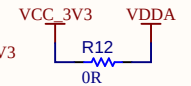
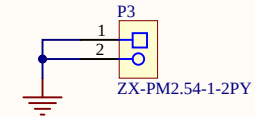
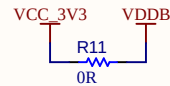
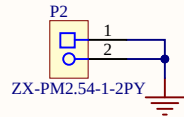
Rev.B-01

Tel qu'assemblé par JLC PCB

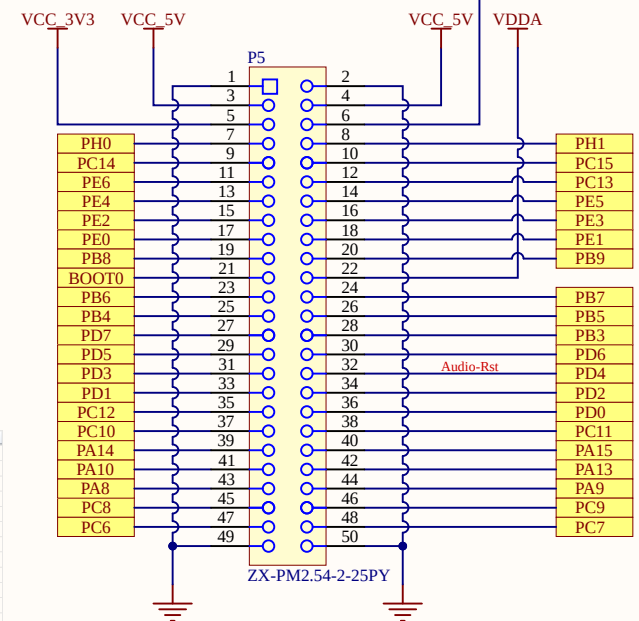
Rev.B-02

Short D2 et D5
Remplacer R10, 0R par 5k1

STM32 DISCOVERY 407



Position	Name	Type	Signal	Label
23	PA0-WKUP	I/O	GPIO_EXTIO	B1 [Blue PushButton]
69	PA10	I/O	USB_OTG_FS_ID	OTG_FS_ID
70	PA11	I/O	USB_OTG_FS_DM	OTG_FS_DM
71	PA12	I/O	USB_OTG_FS_DP	OTG_FS_DP
72	PA13	I/O	SYS_JTMS-SWDIO	SWDIO
76	PA14	I/O	SYS_JTCK-SWCLK	SWCLK
29	PA4	I/O	I2S3_WS	I2S3_WS[CS43L22_LRCK]
30	PA5	I/O	SP11_SCK	SP11_SCK[US302DL_SCLK]
31	PA6	I/O	SP11_MISO	SP11_MISO[US302DL_SDO]
32	PA7	I/O	SP11_MOSI	SP11_MOSI[US302DL_SDASD]
68	PA9	I/O	USB_OTG_FS_VBUS	VBUS_FS
47	PA10	I/O	I2S2_CLK	CLK_IN[MP450T02_CLK]
37	PA2	Input	GPIO_Input	BOOT1
89	PA3	I/O	SYS_JTDO-SWO	SWO
92	PA6	I/O	I2C1_SCL	Audio_SCL[CS43L22_SCL]
96	PA9	I/O	I2C1_SDA	Audio_SDA[CS43L22_SDA]
15	PA0	Output	GPIO_Output	OTG_FS_PowerSwitchOn
78	PA10	I/O	I2S3_CLK	I2S3_SCK[CS43L22_SCLK]
80	PA12	I/O	I2S3_SD	I2S3_SD[CS43L22_SDM]
8	PA14-OSC32_IN	I/O	RCC_OSC32_IN	PC14-OSC32_IN
9	PA15-OSC32_OUT	I/O	RCC_OSC32_OUT	PC15-OSC32_OUT
18	PA3	I/O	I2S2_SD	PD14-OSC32_OUT
64	PA7	I/O	I2S3_MCK	I2S3_MCK[CS43L22_MCLK]
59	PA12	Output	GPIO_Output	LD1 [Green Led]
60	PA13	Output	GPIO_Output	LD3 [Orange Led]
61	PA14	Output	GPIO_Output	LD5 [Red Led]
62	PA15	Output	GPIO_Output	LD6 [Blue Led]
85	PA4	Output	GPIO_Output	Audio_RST[CS43L22_RESET]
86	PA5	Input	GPIO_Input	OTG_FS_OverCurrent
98	PA1	I/O	GPIO_EXTIO	MEMS_INT2[US302DL_INT2]
2	PA3	Output	GPIO_Output	CS120/SP1[US302DL_CS120/SP1]
12	PA0-OSC_IN	I/O	RCC_OSC_IN	PA0-OSC_IN
13	PA1-OSC_OUT	I/O	RCC_OSC_OUT	PA1-OSC_OUT



The JTAG pins are in input PU/PD after reset:
PA15: JTDI in PU
PA14: JTCK in PD
PA13: JTMS in PU
PB4: NJTRST in PU

Titre : STM32 DISCOVERY 407

Conception : CEGEP Limoilou

Approbation : kc

Révision : Rev.B-02

Date : 2026-04-12

Heure : 18:22:34

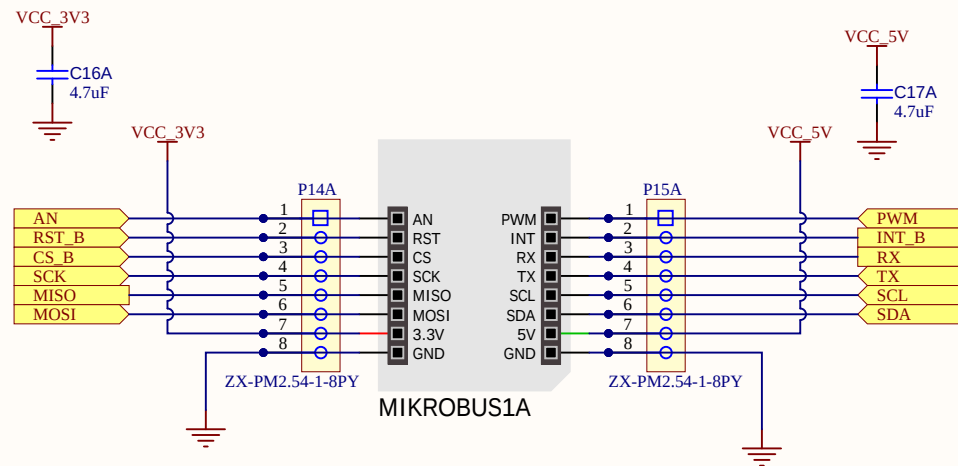
Feuille 2 de 11

Fichier : STM32Disc1-Interface.SchDoc

Cégep Limoilou
1300, 8e Avenue
Québec, QC
Canada
G1J 5L5



Mikro Bus - Connecteur



Titre : Mikro Bus - Connecteur

Conception :
CEGEP Limoilou

Date : 2026-04-12
Fichier : MikroBus-Connecteur.SchDoc

Approbation :
kc

Heure : 18:22:36

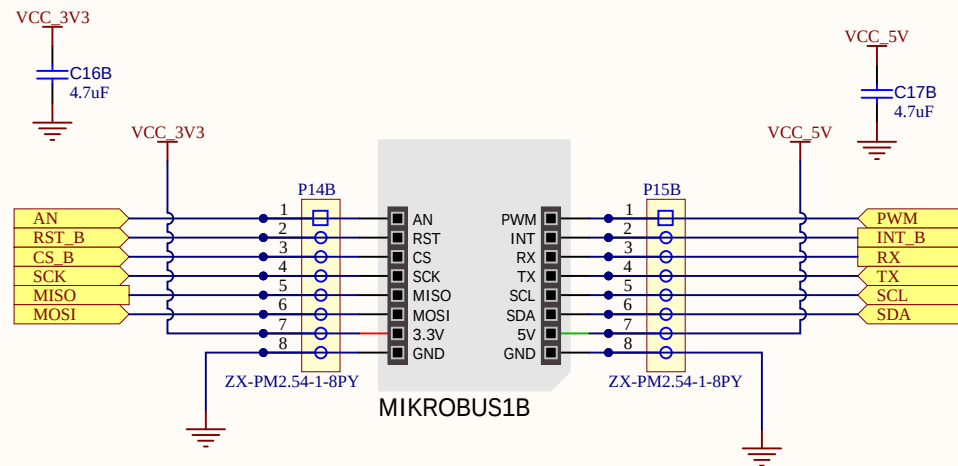
Révision : Rev.B-02

Feuille 3 de 11

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1300, 8e Avenue
Québec, QC
Canada
G1J 5L5



Mikro Bus - Connecteur



Titre : Mikro Bus - Connecteur

Conception :
CEGEP Limoilou

Approbation :
kc

Révision : Rev.B-02

Date : 2026-04-12

Heure : 18:22:37

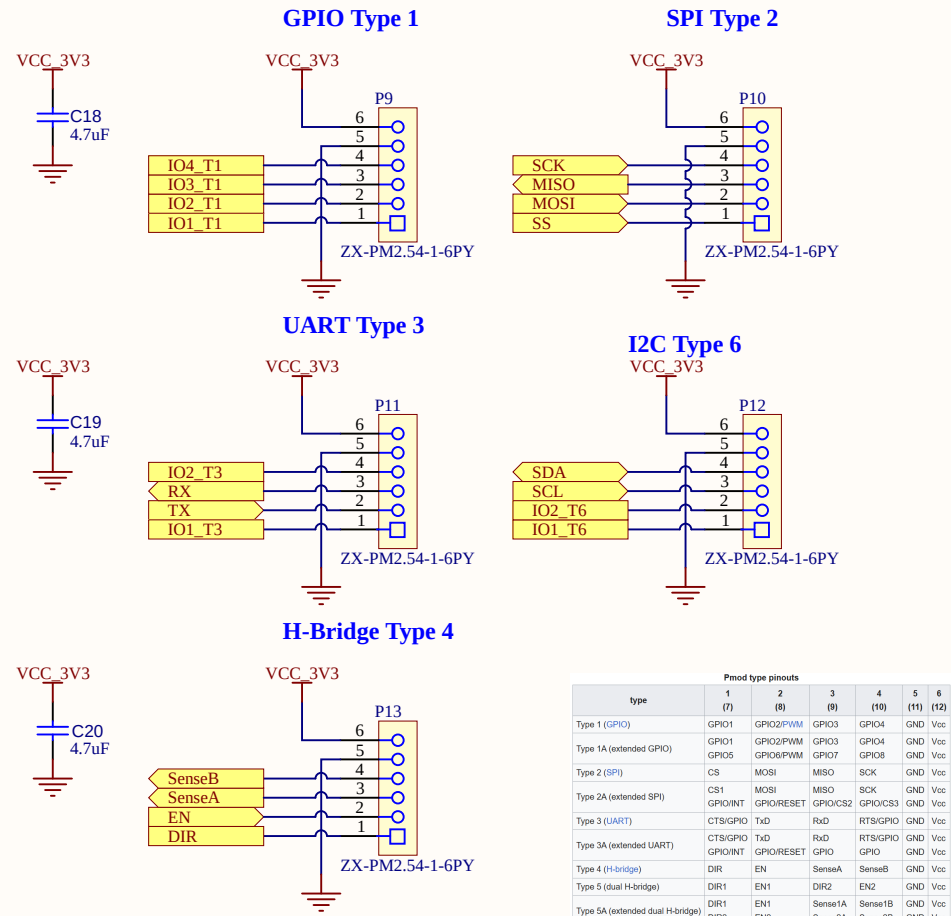
Feuille 3 de 11

Fichier : MikroBus-Connecteur.SchDoc

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Québec, QC
Canada
G1J 5L5

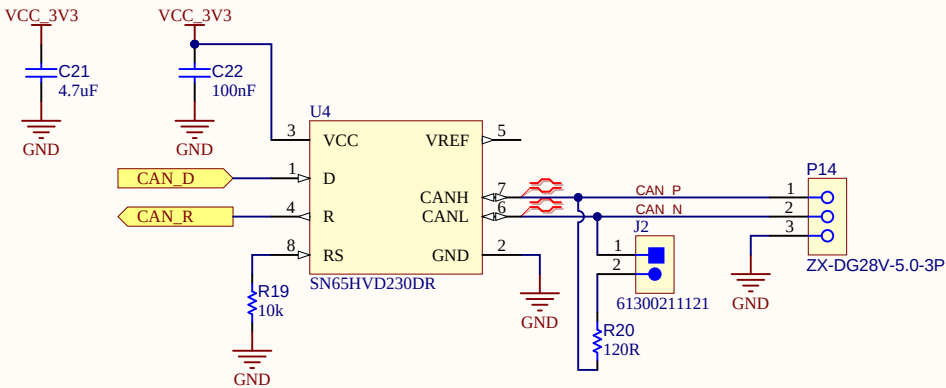


Connecteurs - PMOD Interface



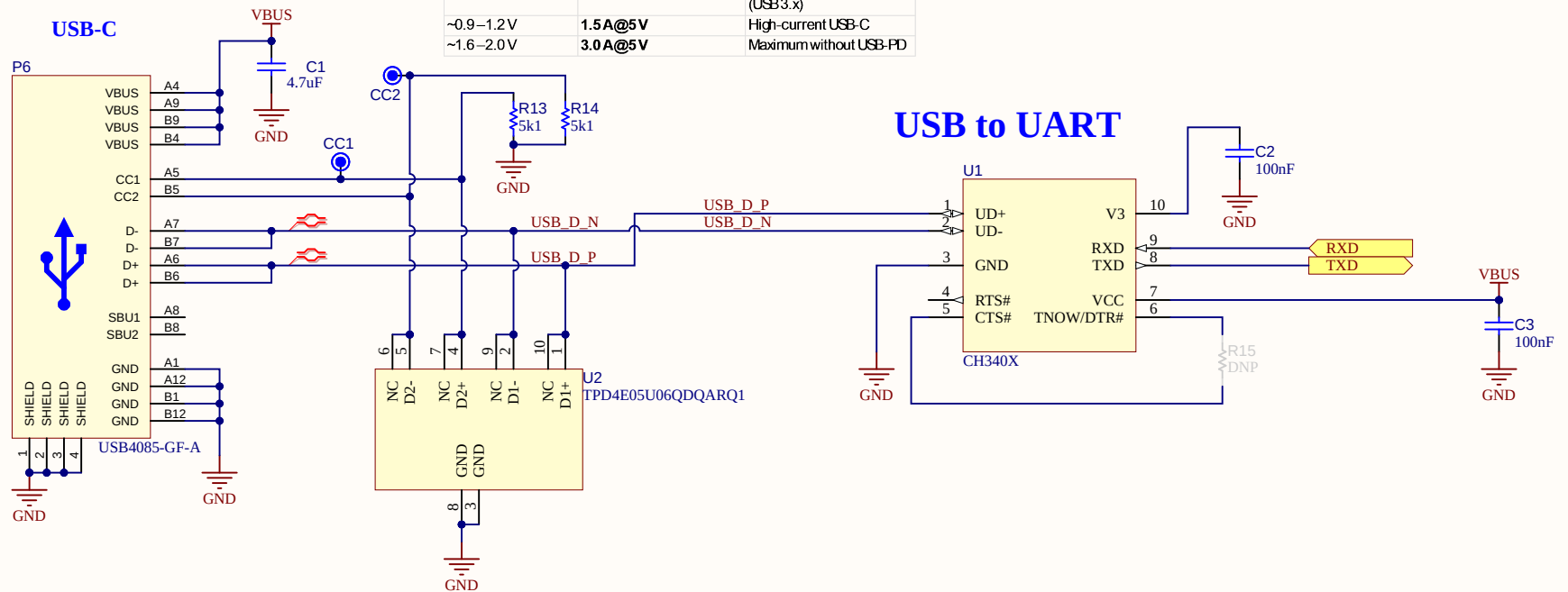
Pmod type pinouts						
type	1 (7)	2 (8)	3 (9)	4 (10)	5 (11)	6 (12)
Type 1 (GPIO)	GPIO1	GPIO2/PWM	GPIO3	GPIO4	GND	Vcc
Type 1A (extended GPIO)	GPIO1	GPIO2/PWM	GPIO3	GPIO4	GND	Vcc
Type 2 (SPI)	CS	MOSI	MISO	SCK	GND	Vcc
Type 2A (extended SPI)	CS1	MOSI	MISO	SCK	GND	Vcc
Type 3 (UART)	CTS/GPIO	TxD	RxD	RTS/GPIO	GND	Vcc
Type 3A (extended UART)	CTS/GPIO	TxD	RxD	RTS/GPIO	GND	Vcc
Type 4 (H-bridge)	DIR	EN	SenseA	SenseB	GND	Vcc
Type 5 (dual H-bridge)	DIR1	EN1	DIR2	EN2	GND	Vcc
Type 5A (extended dual H-bridge)	DIR1	EN1	Sense1A	Sense1B	GND	Vcc
Type 6 (I2C)	nc/INT	nc/RESET	SCL	SDA	GND	Vcc
Type 6A (extended I2C)	nc/INT	nc/RESET	SCL	SDA	GND	Vcc
Type 7 (I2S)	LRCLK	DACdata	ADCLdata	BCLK	GND	Vcc

Driver CAN

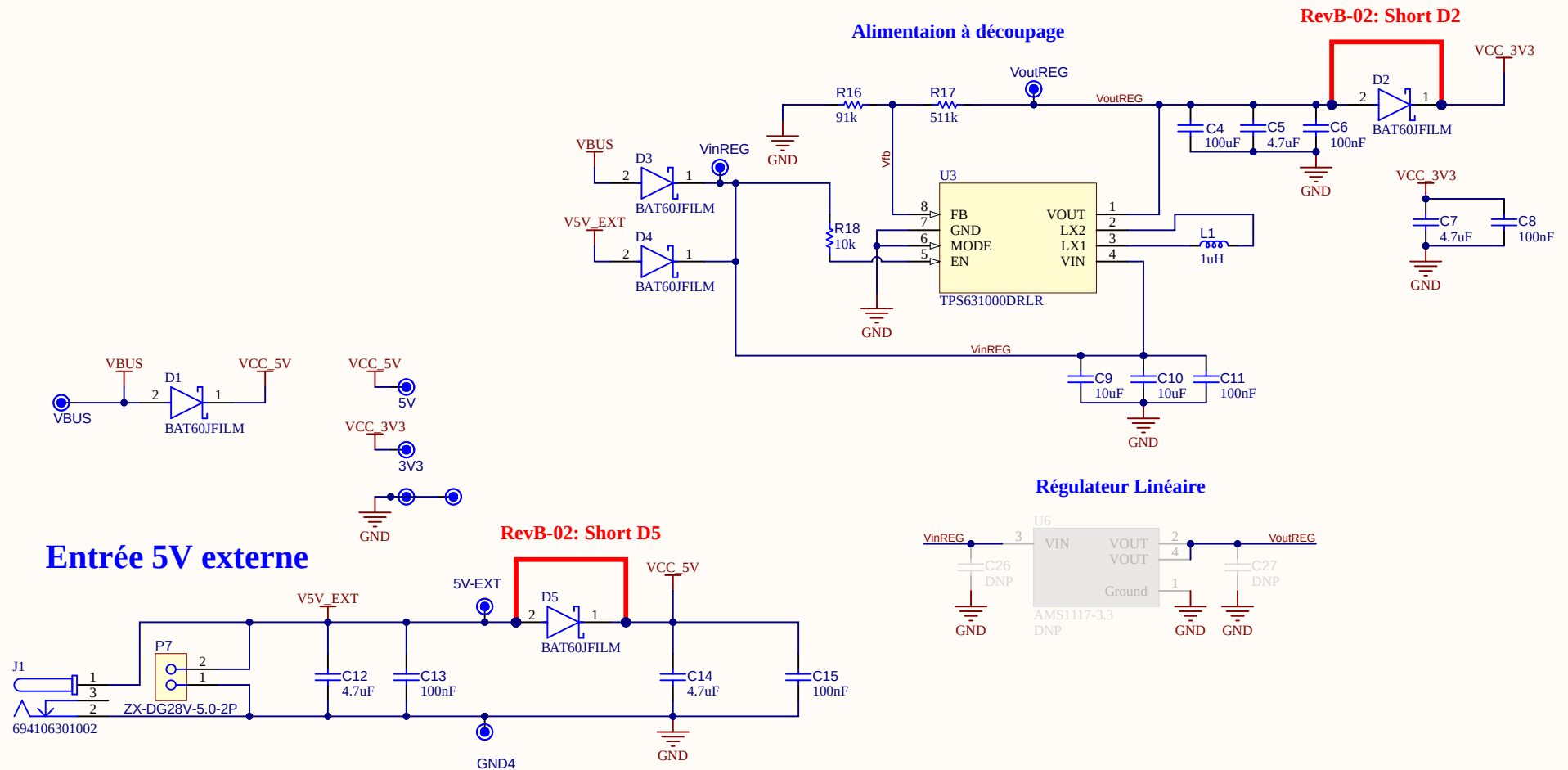


USB to UART

Titre : Interface USB-UART		
Conception : CEGEP Limoilou	Approbation : kc	Révision : Rev.B-02
Date : 2026-04-12	Heure : 18:22:41	Feuille 6 de 11
Fichier : USB.SchDoc		



Alimentation



Titre : Alimentation

Conception :
CEGEP Limoilou

Date : 2026-04-12
Fichier : Alimentation.SchDoc

Approbation :
kc

Heure : 18:22:43

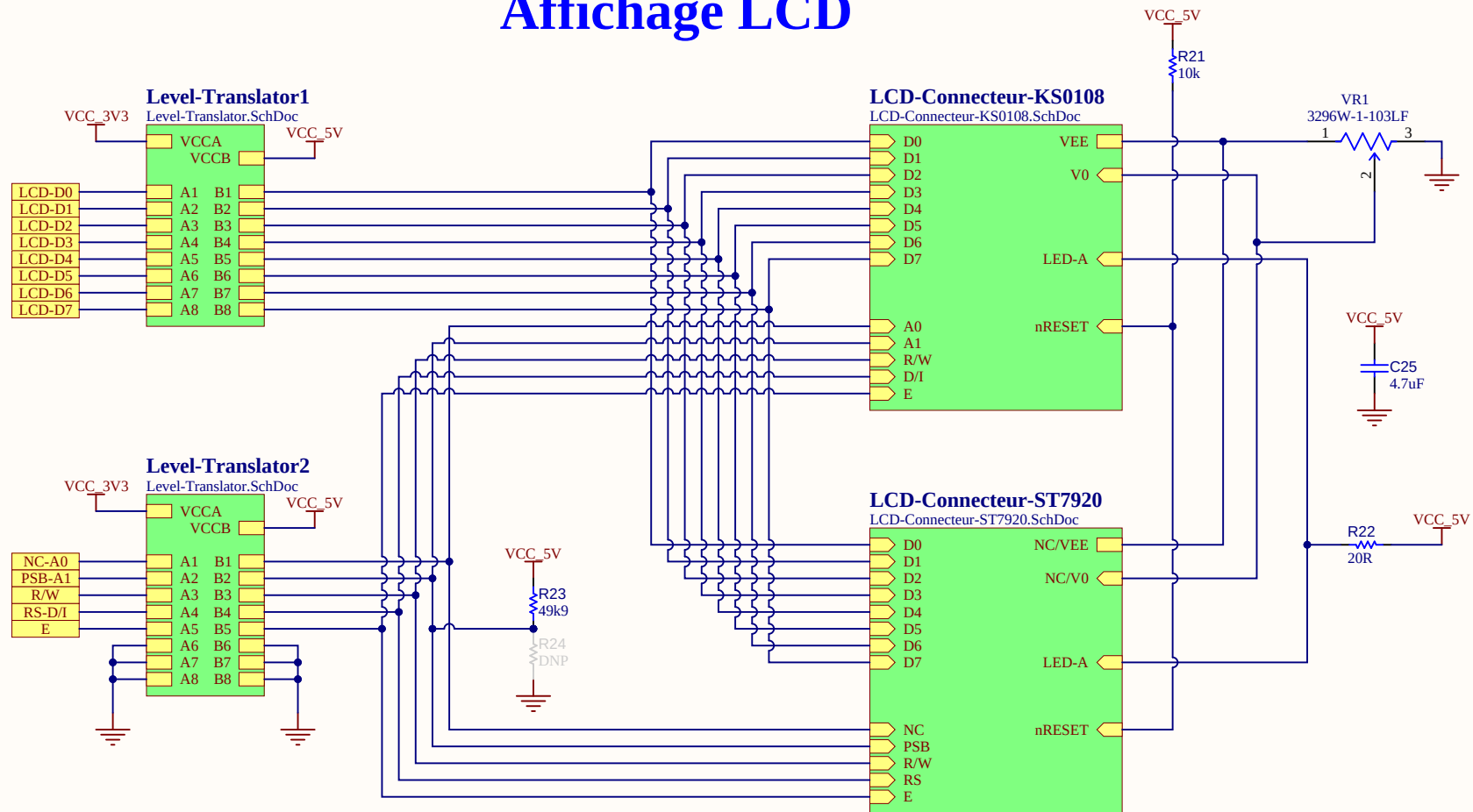
Révision : Rev.B-02

Feuille 7 de 11

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Affichage LCD



Titre : Affichage LCD

Conception :
CEGEP Limoilou

Date : 2026-04-12

Fichier : Affichage-LCD.SchDoc

Approbation :
kc

Heure : 18:22:44

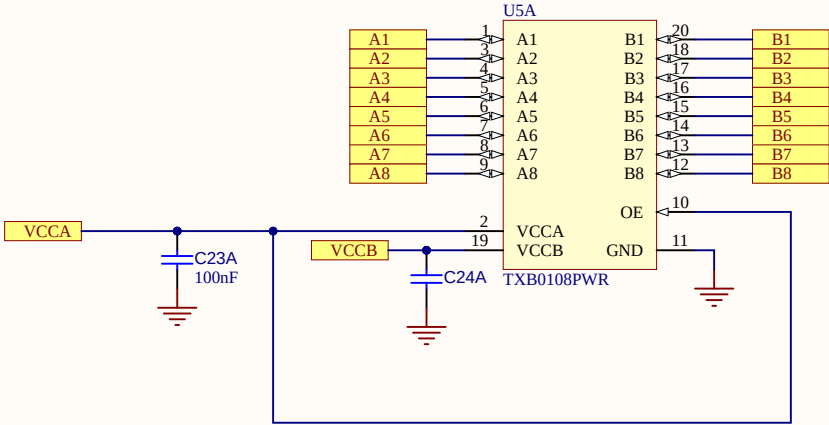
Révision : Rev.B-02

Feuille 8 de 11

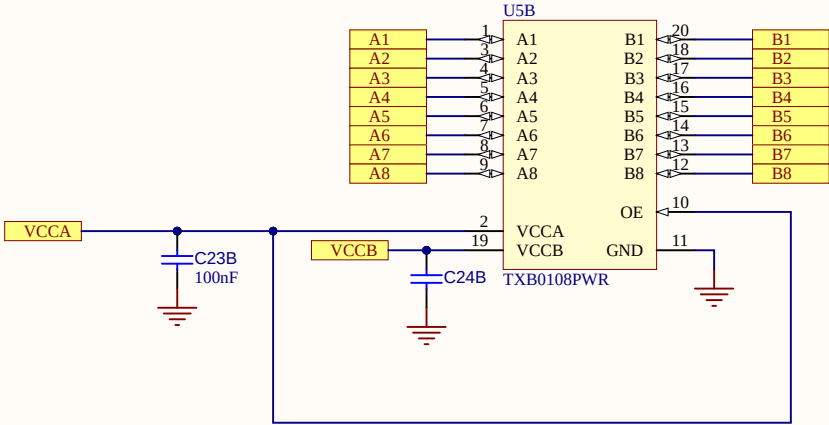
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Canada
G1J 5L5



Bidir Level Translator

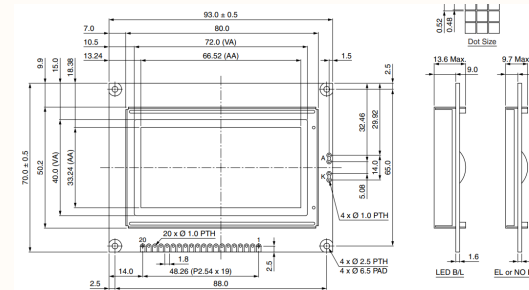
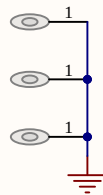
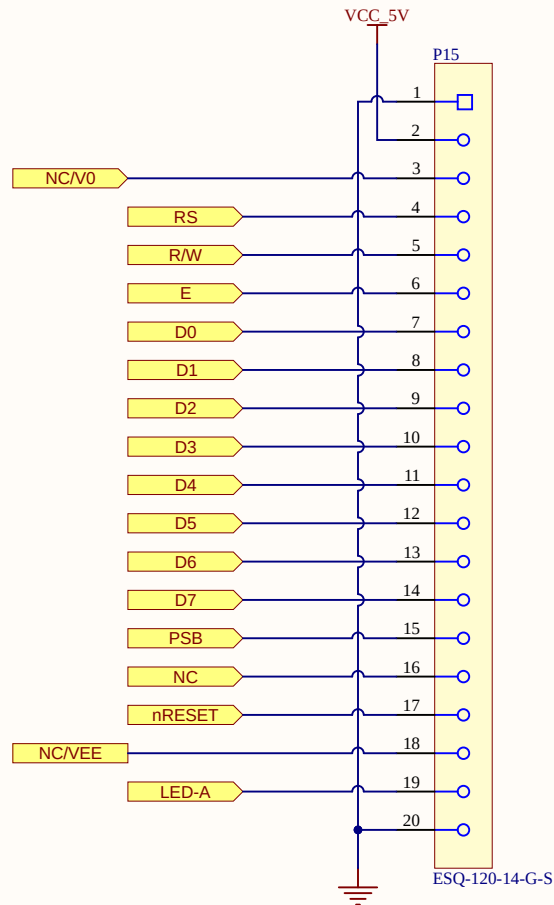


Bidir Level Translator



LCD - Connecteurs

Option LCD 12864 70x93mm



INTERFACE PIN FUNCTION		
PIN NO.	SYMBOL	FUNCTION
1	Vss	Ground
2	VDD	Power supply (+5.0V)
3	Vc	Contrast adjustment
4	DI	Data/Instruction
5	R/W	Data read/write
6	E	H → L enable signal
7	DB0	Data bus line
8	DB1	Data bus line
9	DB2	Data bus line
10	DB3	Data bus line
11	DB4	Data bus line
12	DB5	Data bus line
13	DB6	Data bus line
14	DB7	Data bus line
15	CS1	Chip select for IC1
16	CS2	Chip select for IC2
17	/RES	Reset
18	VEE	Negative voltage output
19	LED+	Power supply for LED (+5.0V, ILED = 0.0)
20	LED-	Power supply for LED (0V)

Pin Description and Wiring Diagram

Pin No.	Symbol	External Connection	Function Description
1	Vss	Power Supply	Ground
2	VDD	Power Supply	Supply Voltage for Logic (+5.0V)
3	Vc	Adj. Power Supply	Supply Voltage for Contrast (approx. -3.7V)
4	RS	MPU	Register Select: 1=Data, 0=Instruction
5	R/W	MPU	Read/Write select signal, R/W=1: Read R/W=0: Write
6	E	MPU	Operation Enable signal. Falling edge triggered.
7-14	DB0-DB7	MPU	This is an 8-bit Bi-directional data bus
15	CS1B	MPU	Chip Selection: CS1=H, CS2=L → select IC1 (left side)
16	CS2B	MPU	CS1=L, CS2=H → select IC2 (right side)
17	/RES	MPU	Active LOW Reset signal
18	VEE	Power Supply	Negative voltage output (-10V)
19	LED+	Power Supply	Backlight Anode (+5V Via On Board Resistor)
20	LED-	Power Supply	Backlight Cathode

Titre : LCD - Connecteurs

Conception :
CEGEP Limoilou

Date : 2026-04-12

Approbation :
kc

Heure : 18:22:48

Révision : Rev.B-02

Feuille 10 de 11

Fichier : LCD-Connecteur-ST7920.SchDoc

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Québec, QC
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LCD - Connecteurs

Option LCD KS0108
50x75mm

